10 Minimum Makespan Scheduling

A central problem in scheduling theory is the following.

Problem 10.1 (Minimum makespan scheduling) Given processing times for n jobs, $p_1, p_2, \ldots, p_n$, and an integer m, find an assignment of the jobs to m identical machines so that the completion time, also called the *makespan*, is minimized.

We will give a simple factor 2 algorithm for this problem before presenting a PTAS for it.

10.1 Factor 2 algorithm

The algorithm is very simple: schedule the jobs one by one, in an arbitrary order, each job being assigned to a machine with least amount of work so far. This algorithm is based on the following two lower bounds on the optimal makespan, OPT:

1. The average time for which a machine has to run, $\left(\sum_i p_i\right)/m$; and
2. The largest processing time.

Let LB denote the combined lower bound, i.e.,

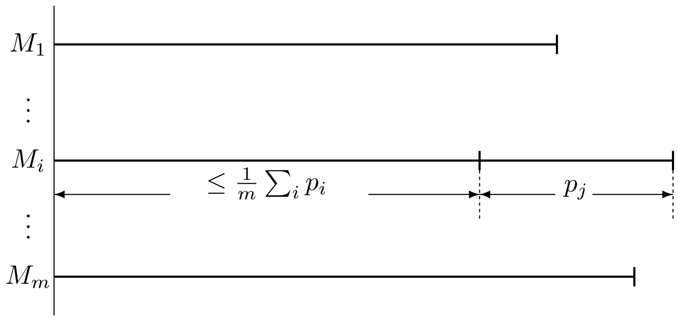
$$\mathrm{LB} = \max\left\{\frac{1}{m}\sum_i p_i,\ \ \max_i\{p_i\}\right\}.$$

Algorithm 10.2 (Minimum makespan scheduling)

1. Order the jobs arbitrarily.
2. Schedule jobs on machines in this order, scheduling the next job on the machine that has been assigned the least amount of work so far.

Theorem 10.3 *Algorithm 10.2 achieves an approximation guarantee of 2 for the minimum makespan problem.*

Proof: Let M_i be the machine that completes its jobs last in the schedule produced by the algorithm, and let j be the index of the last job scheduled on this machine.



Let $start_j$ be the time at which job j starts execution on M_i. Since the algorithm assigns a job to the least loaded machine, it follows that all machines are busy until $start_j$. This implies that

$$start_j \leq \frac{1}{m}\sum_i p_i \leq \mathrm{OPT}.$$

Further, $p_j \leq \mathrm{OPT}$. Thus, the makespan of the schedule is $start_j + p_j \leq 2 \cdot \mathrm{OPT}$. □

Example 10.4 A tight example for this algorithm is provided by a sequence of m^2 jobs with unit processing time, followed by a single job of length m. The schedule obtained by the algorithm has a makespan of $2m$, while $\mathrm{OPT} = m + 1$. □

10.2 A PTAS for minimum makespan

The minimum makespan problem is strongly **NP**-hard; thus, by Corollary 8.6, it does not admit an FPTAS, assuming $\mathbf{P} \neq \mathbf{NP}$. We will obtain a PTAS for it. The minimum makespan problem is closely related to the bin packing problem by the following observation. There exists a schedule with makespan t iff n objects of sizes $p_1, p_2, \ldots, p_n$ can be packed into m bins of capacity t each. This suggests a reduction from minimum makespan to bin packing as follows. Denoting the sizes of the n objects, $p_1, \ldots, p_n$, by I, let $\mathrm{bins}(I, t)$ represent the minimum number of bins of size t required to pack these n objects. Then, the minimum makespan is given by

$$\min\{t : \ \mathrm{bins}(I, t) \leq m\}.$$